



Page 1 of 7 · A Sketch the Size of a Postcard

The town square has one long blank wall, and the Harvest Festival is in three days. Vincent Van Mouse unrolls his sketch: a sunset over the hills, small enough to fit in his paw. “Every centimetre on my sketch becomes 25 centimetres on the wall,” he explains. Stripe-Scout whistles. Sail-Affe stretches her neck to see the top of the wall.



Page 2 of 7 · Sunset Orange

Vincent’s sunset orange is always mixed from red and yellow in the ratio 2:3.
Stripe-Scout has already poured 8 pots of red into the big mixing tub and is waiting, brush ready, for the yellow.



Page 3 of 7 · The Wrong Shade?

On the second morning, Stripe-Scout finds yesterday’s tub. Its label says 10 pots of red to 15 pots of yellow. Her ears flatten. “I mixed it wrong. It’s ruined.” Vincent doesn’t answer straight away. He simplifies the ratio first, the way a careful artist checks a colour before judging it.



Page 4 of 7 · High Squares and Low Squares

Vincent chalks a grid of 45 squares onto the wall. Sail-Affe can reach the high squares without a ladder, and Stripe-Scout is quickest along the bottom. They agree to split the squares in the ratio 4:5, high to low.



Page 5 of 7 · Racing the Rain Clouds

On the last afternoon, grey clouds roll over the hills. Seven squares are still bare. Stripe-Scout wants to give up and throw a sheet over the wall, but Vincent checks the clock instead. They have been painting 3 squares every 12 minutes, steady and neat.



Page 6 of 7 · Gold for the Final Touch

They finish with minutes to spare, and the first raindrops miss the wall. One small pot of gold paint is left for the sun’s rays. Vincent shares its 30 ml in the ratio 1:2:3 between himself, Sail-Affe and Stripe-Scout, because Stripe-Scout has the most rays to paint.



Page 7 of 7 · The Unveiling

At the festival, the whole town gathers in front of the wall. The sunset glows exactly like the sketch, only twenty-five times taller. Professor Hooten asks for a postcard of the mural for every reader in the library. Vincent smiles. Scaling works in both directions.

Questions for Vincent Van Mouse and the Mural in Proportion

Read the story together first. Use one question at each page, then the question at the end. Answers are below the dotted line: fold them under or cover them while reading.

Page 1 · A Sketch the Size of a Postcard

The sketch is 8 cm tall. If every 1 cm becomes 25 cm, how tall will the mural be, in centimetres?

Choices: 33 · 175 · 200 · 225

Page 2 · Sunset Orange

Red and yellow are mixed in the ratio 2:3. There are 8 pots of red. How many pots of yellow are needed?

Choices: 9 · 12 · 16 · 24

Page 3 · The Wrong Shade?

Simplify 10:15 so that the first number is 2. What is the second number?

Choices: 3 · 5 · 7 · 13

Page 4 · High Squares and Low Squares

45 squares are shared in the ratio 4:5. How many squares does Sail-Affe paint?

Choices: 9 · 20 · 25 · 36

Page 5 · Racing the Rain Clouds

At the same rate, how many minutes will 7 squares take?

Choices: 16 · 21 · 28 · 84

Page 6 · Gold for the Final Touch

30 ml is shared in the ratio 1:2:3. How many millilitres does Stripe-Scout get?

Choices: 3 · 5 · 10 · 15

Page 7 · The Unveiling

The mural is 200 cm tall. At a scale of 1 to 25, how tall is the postcard picture, in centimetres?

Choices: 8 · 175 · 225 · 5000

Answers (fold under before reading)

Page 1: 200. Scaling multiplies every length by the same factor: 8 x 25 = 200 cm. Adding 25 instead (8 + 25 = 33) would squash the picture out of proportion.

Page 2: 12. 2 parts are 8 pots, so 1 part is 4 pots. Yellow is 3 parts: 3 x 4 = 12. Keeping the same difference instead (8 + 1 = 9) would change the colour.

Page 3: 3. Divide both parts by 5: 10 ÷ 5 = 2 and 15 ÷ 5 = 3, so 10:15 is 2:3, the very same orange. Taking 8 away from both parts (2:7) does not keep the ratio.

Page 4: 20. 4 + 5 = 9 parts altogether. 45 ÷ 9 = 5 squares in each part. Sail-Affe paints 4 parts: 4 x 5 = 20.

Page 5: 28. Find the time for one square first: 12 ÷ 3 = 4 minutes. Then 7 x 4 = 28 minutes. Adding 4 more minutes (12 + 4 = 16) treats a proportion as an addition.

Page 6: 15. 1 + 2 + 3 = 6 parts. 30 ÷ 6 = 5 ml in each part. Stripe-Scout has 3 parts: 3 x 5 = 15 ml.

Page 7: 8. To scale down, divide every length by 25: 200 ÷ 25 = 8 cm. That is exactly the size of Vincent’s first sketch.